MINIMUM SPANNING TREE

ALGORITHMS

1. KRUSKAL'S ALGORITHM
2. PRIM'S ALGORITHM

Minimum Spanning Tree - Definition

- Given a connected and undirected graph, G (V, E), a spanning tree of that graph is a subgraph that is a tree and connects all the vertices together.

- A single graph can have many different spanning trees.

- A minimum spanning tree (MST) or minimum weight spanning tree for a weighted, connected and undirected graph is a spanning tree with weight less than or equal to the weight of every other spanning tree.

- The weight of a spanning tree is the sum of weights given to each edge of the spanning tree.

- A minimum spanning tree has (V – 1) edges where V is the number of vertices in the given graph.

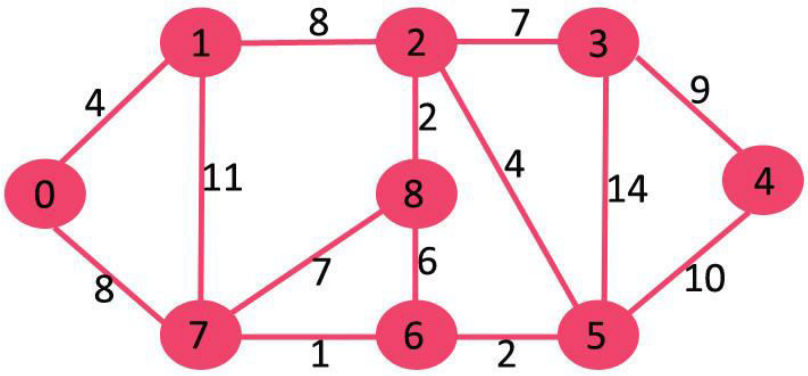
Kruskal's algorithm

Steps for finding MST

1. Sort all the edges in **non-decreasing (increasing) order of their weight.**
2. Pick **the smallest edge**. Check if it forms a cycle with the spanning tree formed so far. **If cycle is not formed, include this edge**. Else, discard it.
3. Repeat **step#2 until there are (V-1) edges** in the spanning tree.

EXAMPLE

- The algorithm is a Greedy Algorithm.
- The **Greedy Choice** is to **pick the smallest weight edge** that does not form a cycle in the MST constructed so far.



- The graph contains 9 vertices and 14 edges. So, the minimum spanning tree formed will be having (9 – 1) = 8 edges

After sorting:

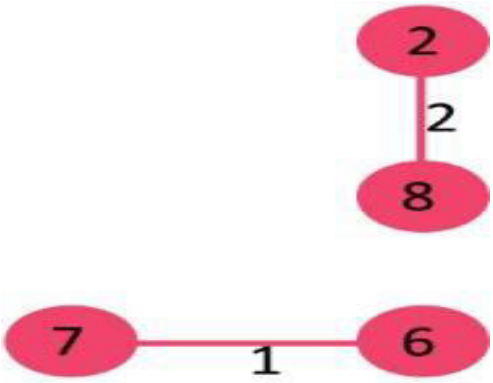
Weight	Source	Dest
1	7	6
2	8	2
2	6	5
4	0	1
4	2	5
6	8	6
7	2	3
7	7	8
8	0	7
8	1	2
9	3	4
10	5	4
11	1	7
14	3	5

Now pick all edges one by one from sorted list of edges

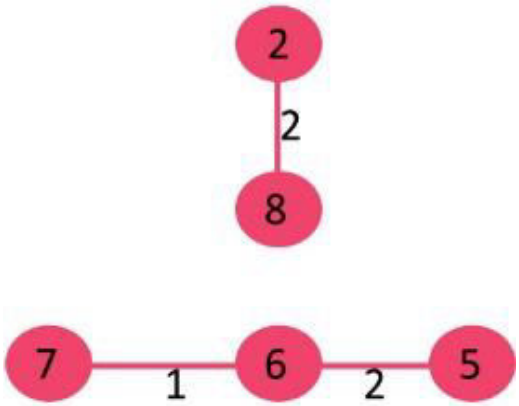
1. *Pick edge 7-6:* No cycle is formed, include it.



2. *Pick edge 8-2:* No cycle is formed, include it.



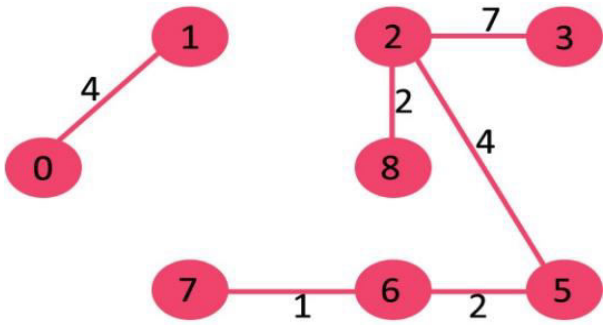
3. *Pick edge 6-5:* No cycle is formed, include it.



Steps 4.& 5 ?

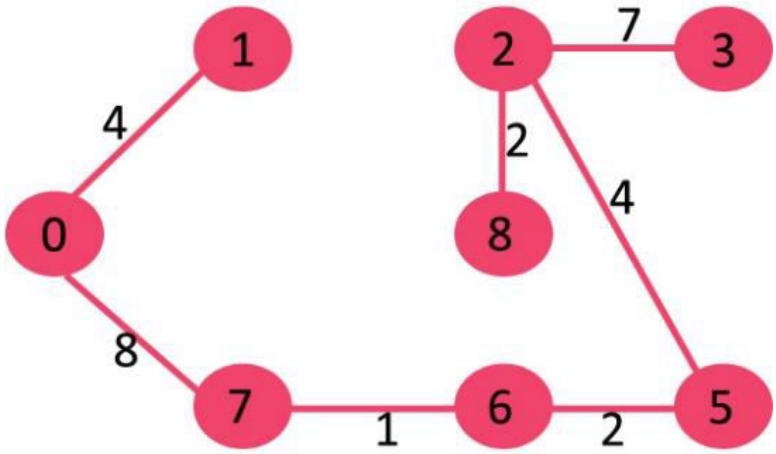
6. *Pick edge 8-6:* Since including this edge results in cycle, discard it.

7. *Pick edge 2-3:* No cycle is formed, include it.



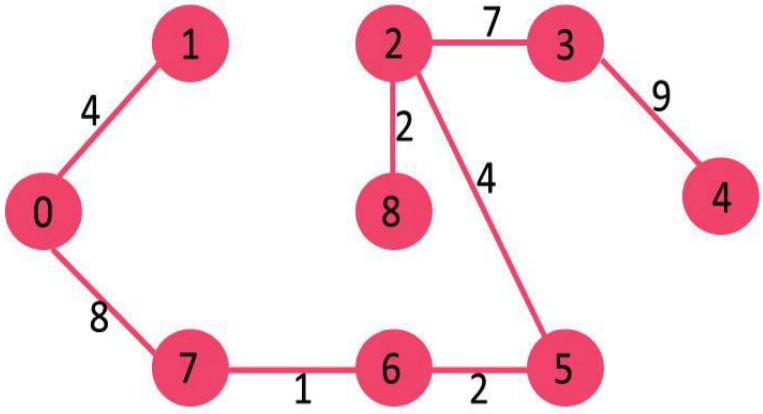
8. Pick edge 7-8: Since including this edge results in cycle, discard it.

9. *Pick edge 0-7:* No cycle is formed, include it.



10. *Pick edge 1-2:* Since including this edge results in cycle, discard it.

11. *Pick edge 3-4:* No cycle is formed, include it.



- Since the number of edges included equals (V – 1), the algorithm stops

Time Complexity: O(E log E) or O(E log V).